K-mer MLP: Baseline, Regularization, and Hyperparameter Tuning

Notebook goal

This notebook is organized around this week's focus: **regularization strategies and hyperparameter tuning** for a multi-label MLP on k-mer protein features.

Design choice

To keep the notebook focused, the early sections only do the minimum setup needed to verify that the pipeline works:

- load processed k-mer features and GO labels
- define reusable datasets, models, and helper functions
- run a short baseline sanity check

The full threshold-based evaluation is deferred to the later tuning sections, where we compare settings systematically.

Task summary

- **Input:** normalized 2-mer frequency vectors, dimension = 400
- **Output:** full GO label space, dimension = 31,454
- **Task type:** multi-label classification
- **This week's emphasis:** generalization, overfitting control, and parameter selection

Step 0. Imports and Data Loading

Why this step?

We begin by loading the processed k-mer features and GO labels that were already prepared in the preprocessing notebook.
This avoids repeating sequence cleaning, k-mer extraction, feature scaling, and GO alignment.

What we load

- `X_train.npy`, `X_val.npy`: standardized k-mer feature matrices
- `Y_train.npz`, `Y_val.npz`: sparse GO target matrices
- `meta.pkl`: metadata such as GO encoders and protein IDs

Why this section is kept short

This notebook is not primarily about preprocessing.
The goal here is simply to bring the prepared data into memory so that the tuning experiments can begin.

In [1]:
```
import os
import json
import copy
import pickle
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from scipy.sparse import load_npz

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader

from sklearn.metrics import f1_score
```

In [2]:
```
DATA_DIR = "data_processed_kmer"

X_train = np.load(os.path.join(DATA_DIR, "X_train.npy"))
X_val   = np.load(os.path.join(DATA_DIR, "X_val.npy"))

Y_train = load_npz(os.path.join(DATA_DIR, "Y_train.npz"))
Y_val   = load_npz(os.path.join(DATA_DIR, "Y_val.npz"))

with open(os.path.join(DATA_DIR, "meta.pkl"), "rb") as f:
    meta = pickle.load(f)

print("X_train:", X_train.shape, X_train.dtype)
print("X_val:  ", X_val.shape, X_val.dtype)
print("Y_train:", Y_train.shape, type(Y_train))
print("Y_val:  ", Y_val.shape, type(Y_val))
```

```
X_train: (112455, 400) float32
X_val:   (28114, 400) float32
Y_train: (112455, 31454) <class 'scipy.sparse._csr.csr_matrix'>
Y_val:   (28114, 31454) <class 'scipy.sparse._csr.csr_matrix'>
```

Step 1. Define Shared Training Components

Why this step?

All later experiments should reuse the same core pipeline.
So instead of rewriting code for every tuning run, we define the reusable building blocks once:

- dataset class
- model class
- loss setup
- training loop
- validation loop
- threshold sweep
- helper functions for experiment configuration

Why this matters

This structure keeps the notebook clean and makes it easier to compare:

- learning rates
- dropout levels
- weight decay values
- final model settings

Important note

These definitions are shared by both the short baseline sanity check and the later regularization/tuning experiments.

In [3]:
```
class ProteinKmerDataset(Dataset):
    def __init__(self, X, Y_sparse):
        self.X = X.astype(np.float32)
        self.Y = Y_sparse.tocsr()
        self.n_labels = self.Y.shape[1]

    def __len__(self):
        return self.X.shape[0]

    def __getitem__(self, idx):
        x = torch.from_numpy(self.X[idx])

        start = self.Y.indptr[idx]
        end = self.Y.indptr[idx + 1]
        pos_idx = self.Y.indices[start:end]

        y = torch.zeros(self.n_labels, dtype=torch.float32)
        y[pos_idx] = 1.0
        return x, y
```

In [4]:
```
train_dataset = ProteinKmerDataset(X_train, Y_train)
val_dataset   = ProteinKmerDataset(X_val, Y_val)

print("Train samples:", len(train_dataset))
print("Val samples:  ", len(val_dataset))
```

```
Train samples: 112455
Val samples:   28114
```

In [5]:
```
class MLP(nn.Module):
    def __init__(self, input_dim, output_dim, hidden1=512, hidden2=256, drop
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden1),
            nn.BatchNorm1d(hidden1),
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden1, hidden2),
            nn.BatchNorm1d(hidden2),
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden2, output_dim)
        )

    def forward(self, x):
        return self.net(x)
```

In [6]:
```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
input_dim = X_train.shape[1]
output_dim = Y_train.shape[1]

print("Device:", device)
print("Input dim:", input_dim)
print("Output dim:", output_dim)
```

```
Device: cpu
Input dim: 400
Output dim: 31454
```

```
/global/home/users/yushanfu/esm_env/lib/python3.11/site-packages/torch/cuda/
__init__.py:180: UserWarning: CUDA initialization: The NVIDIA driver on your
system is too old (found version 12030). Please update your GPU driver by do
wnloading and installing a new version from the URL: http://www.nvidia.com/D
ownload/index.aspx Alternatively, go to: https://pytorch.org to install a Py
Torch version that has been compiled with your version of the CUDA driver.
(Triggered internally at /pytorch/c10/cuda/CUDAFunctions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
```

In [7]:
```
label_counts = np.asarray(Y_train.sum(axis=0)).ravel().astype(np.float32)
n_train = Y_train.shape[0]

# Compute positive class weights for each GO term.
# Add epsilon to avoid division by zero, then clip extreme values for stabil
pos_weight = (n_train - label_counts) / (label_counts + 1e-6)
pos_weight = np.clip(pos_weight, 1.0, 100.0)

pos_weight_tensor = torch.tensor(pos_weight, dtype=torch.float32, device=dev
criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight_tensor)

print("label count min/max:", label_counts.min(), label_counts.max())
print("pos_weight min/max:", pos_weight.min(), pos_weight.max())
```

```
label count min/max: 0.0 73516.0
pos_weight min/max: 1.0 100.0
```

In [8]:
```
def build_dataloaders(train_dataset, val_dataset, batch_size):
    train_loader = DataLoader(
        train_dataset,
        batch_size=batch_size,
        shuffle=True,
        num_workers=4,
        pin_memory=torch.cuda.is_available()
    )

    val_loader = DataLoader(
        val_dataset,
        batch_size=batch_size,
        shuffle=False,
        num_workers=4,
        pin_memory=torch.cuda.is_available()
    )
    return train_loader, val_loader


def build_model(config, input_dim, output_dim, device):
    return MLP(
        input_dim=input_dim,
        output_dim=output_dim,
        hidden1=config["hidden1"],
        hidden2=config["hidden2"],
        dropout=config["dropout"]
    ).to(device)


def build_optimizer(model, config):
    return optim.Adam(
        model.parameters(),
        lr=config["learning_rate"],
        weight_decay=config["weight_decay"]
    )
```

In [9]:
```
def train_one_epoch(model, loader, criterion, optimizer, device):
    model.train()
    total_loss = 0.0

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        Y_batch = Y_batch.to(device, non_blocking=True)

        optimizer.zero_grad()
        logits = model(X_batch)
        loss = criterion(logits, Y_batch)
        loss.backward()
        optimizer.step()

        total_loss += loss.item()

    return total_loss / len(loader)
```

```
@torch.no_grad()
def evaluate_loss(model, loader, criterion, device):
    model.eval()
    total_loss = 0.0

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        Y_batch = Y_batch.to(device, non_blocking=True)

        logits = model(X_batch)
        loss = criterion(logits, Y_batch)

        total_loss += loss.item()

    return total_loss / len(loader)


@torch.no_grad()
def get_predictions(model, loader, device):
    model.eval()
    all_probs = []
    all_targets = []

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        logits = model(X_batch)
        probs = torch.sigmoid(logits).cpu().numpy()

        all_probs.append(probs)
        all_targets.append(Y_batch.numpy())

    all_probs = np.vstack(all_probs)
    all_targets = np.vstack(all_targets)
    return all_probs, all_targets
```

In [10]:

```
def threshold_sweep(y_true, y_prob, thresholds=None):
    if thresholds is None:
        thresholds = np.arange(0.05, 0.95, 0.1)

    records = []
    for t in thresholds:
        y_pred = (y_prob >= t).astype(np.int8)

        micro = f1_score(y_true, y_pred, average="micro", zero_division=0)
        macro = f1_score(y_true, y_pred, average="macro", zero_division=0)

        records.append({
            "threshold": float(t),
            "micro_f1": float(micro),
            "macro_f1": float(macro)
        })

    best_micro_row = max(records, key=lambda x: x["micro_f1"])
    best_macro_row = max(records, key=lambda x: x["macro_f1"])
```

```
    return records, best_micro_row, best_macro_row


def train_with_early_stopping(model, train_loader, val_loader, criterion, op
                              num_epochs=10, patience=3, save_path=None):
    best_val_loss = float("inf")
    best_state_dict = None
    epochs_no_improve = 0

    train_losses = []
    val_losses = []

    for epoch in range(num_epochs):
        train_loss = train_one_epoch(model, train_loader, criterion, optimiz
        val_loss = evaluate_loss(model, val_loader, criterion, device)

        train_losses.append(train_loss)
        val_losses.append(val_loss)

        print(f"Epoch {epoch+1:02d}/{num_epochs} | train_loss={train_loss:.6

        if val_loss < best_val_loss:
            best_val_loss = val_loss
            best_state_dict = copy.deepcopy(model.state_dict())
            epochs_no_improve = 0

            if save_path is not None:
                torch.save(best_state_dict, save_path)
        else:
            epochs_no_improve += 1

        if epochs_no_improve >= patience:
            print(f"Early stopping triggered at epoch {epoch+1}")
            break

    if best_state_dict is not None:
        model.load_state_dict(best_state_dict)

    return model, train_losses, val_losses, best_val_loss


def run_experiment(config, train_dataset, val_dataset, input_dim, output_dim
                   run_threshold_sweep=True):
    train_loader, val_loader = build_dataloaders(
        train_dataset=train_dataset,
        val_dataset=val_dataset,
        batch_size=config["batch_size"]
    )

    model = build_model(config, input_dim, output_dim, device)
    optimizer = build_optimizer(model, config)
    save_path = f"{config['save_prefix']}.pt"

    model, train_losses, val_losses, best_val_loss = train_with_early_stoppi
        model=model,
        train_loader=train_loader,
```

```
        val_loader=val_loader,
        criterion=criterion,
        optimizer=optimizer,
        device=device,
        num_epochs=config["num_epochs"],
        patience=config["patience"],
        save_path=save_path
    )

    result = {
        "config": config.copy(),
        "model": model,
        "model_path": save_path,
        "best_val_loss": best_val_loss,
        "train_losses": train_losses,
        "val_losses": val_losses,
    }

    if run_threshold_sweep:
        val_probs, val_targets = get_predictions(model, val_loader, device)
        records, best_micro_row, best_macro_row = threshold_sweep(val_target
        result["threshold_records"] = records
        result["best_micro_row"] = best_micro_row
        result["best_macro_row"] = best_macro_row

    return result
```

In [11]:
```
np.random.seed(42)

small_train_size = len(train_dataset) // 10
small_val_size = len(val_dataset) // 10

small_train_idx = np.random.choice(len(train_dataset), size=small_train_size
small_val_idx = np.random.choice(len(val_dataset), size=small_val_size, repl
```

In [12]:
```
print(small_train_size)
```

```
11245
```

In [13]:
```
from torch.utils.data import Subset

train_dataset_small = Subset(train_dataset, small_train_idx)
val_dataset_small = Subset(val_dataset, small_val_idx)
```

Step 2. Quick Baseline Sanity Check

Why this step?

Before running full tuning experiments, we perform a short baseline run to verify that:

- the model trains without numerical issues
- validation loss stays finite
- checkpointing and dataloaders behave correctly

Why we keep it short

This section is **not** the main focus of the notebook.
It only serves as a sanity check, so we use a small number of epochs.

Important note

We intentionally do **not** run threshold-based F1 evaluation here.
Threshold selection is more meaningful in the later tuning and regularization sections, where settings are compared systematically.

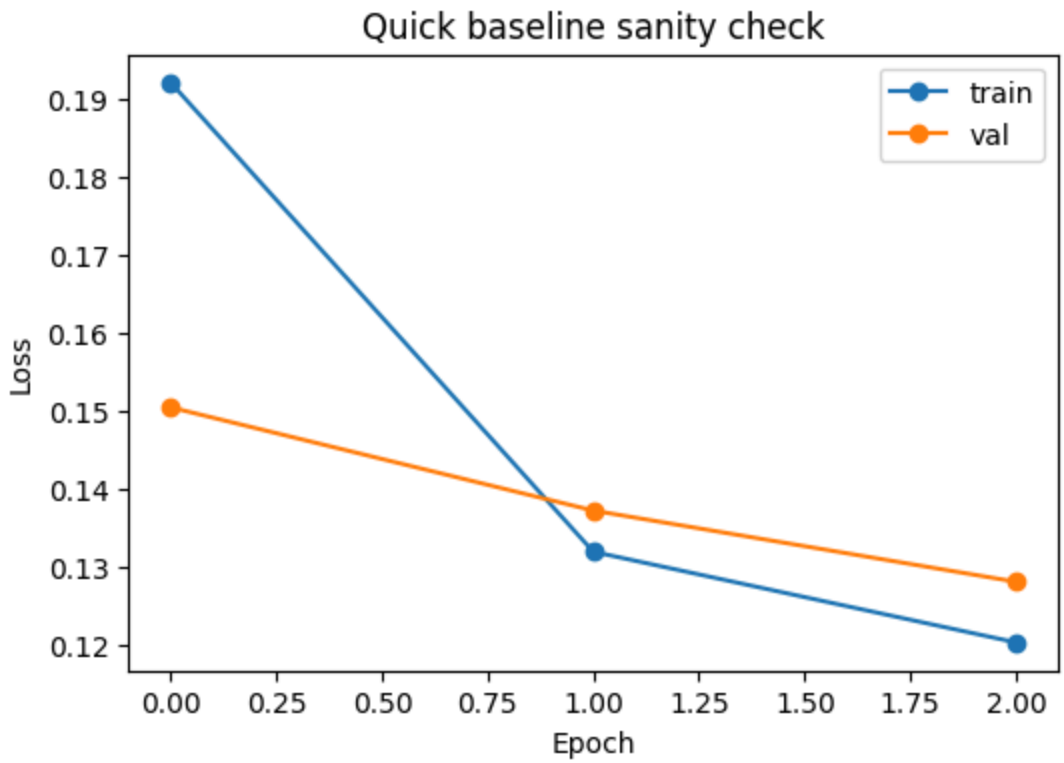
In [14]:
```
BASELINE_CONFIG = {
    "hidden1": 512,
    "hidden2": 256,
    "dropout": 0.3,
    "learning_rate": 1e-3,
    "weight_decay": 1e-5,
    "batch_size": 64,
    "num_epochs": 3,
    "patience": 2,
    "save_prefix": "baseline_sanity_check"
}

baseline_result = run_experiment(
    config=BASELINE_CONFIG,
    train_dataset=train_dataset_small,
    val_dataset=val_dataset_small,
    input_dim=input_dim,
    output_dim=output_dim,
    criterion=criterion,
    device=device,
    run_threshold_sweep=False
)

print("Baseline best val loss:", baseline_result["best_val_loss"])
```

```
Epoch 01/3 | train_loss=0.192200 | val_loss=0.150537
Epoch 02/3 | train_loss=0.132022 | val_loss=0.137304
Epoch 03/3 | train_loss=0.120395 | val_loss=0.128172
Baseline best val loss: 0.12817196361720562
```

In [15]:
```
plt.figure(figsize=(6, 4))
plt.plot(baseline_result["train_losses"], marker="o", label="train")
plt.plot(baseline_result["val_losses"], marker="o", label="val")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Quick baseline sanity check")
plt.legend()
plt.show()
```



Step 3. Hyperparameter Tuning: Learning Rate

Why this step?

We tune the learning rate first because it strongly affects optimization stability and convergence speed.

What is fixed in this section?

We keep the architecture and regularization settings fixed, and only vary:

- `learning_rate`

Selection rule

For each run, we perform threshold sweeping on the validation set and select the best learning rate using **best micro-F1**.

In [16]:
```
TUNING_BASE_CONFIG = {
    "hidden1": 512,
    "hidden2": 256,
    "dropout": 0.3,
    "learning_rate": 1e-3,
    "weight_decay": 1e-5,
    "batch_size": 64,
    "num_epochs": 3,
    "patience": 3,
```

```
    "save_prefix": "tuning_run"
}
```

In [17]:
```
learning_rates = [1e-2, 1e-3, 1e-4]
lr_results = []

for lr in learning_rates:
    print("=" * 80)
    print(f"Learning rate tuning: lr = {lr}")
    print("=" * 80)

    config = TUNING_BASE_CONFIG.copy()
    config["learning_rate"] = lr
    config["save_prefix"] = f"kmer_lr_{lr}"

    result = run_experiment(
        config=config,
        train_dataset=train_dataset_small,
        val_dataset=val_dataset_small,
        input_dim=input_dim,
        output_dim=output_dim,
        criterion=criterion,
        device=device,
        run_threshold_sweep=True
    )

    lr_results.append({
        "learning_rate": lr,
        "best_val_loss": result["best_val_loss"],
        "best_micro_f1": result["best_micro_row"]["micro_f1"],
        "best_macro_f1": result["best_macro_row"]["macro_f1"],
        "best_threshold_micro": result["best_micro_row"]["threshold"],
        "best_threshold_macro": result["best_macro_row"]["threshold"],
        "model_path": result["model_path"]
    })

df_lr = pd.DataFrame(lr_results).sort_values("best_micro_f1", ascending=Fals
display(df_lr)
```

```
==============================================================================
====
Learning rate tuning: lr = 0.01
==============================================================================
====
Epoch 01/3 | train_loss=0.167909 | val_loss=0.149168
Epoch 02/3 | train_loss=0.149745 | val_loss=0.146881
Epoch 03/3 | train_loss=0.143525 | val_loss=0.143856
==============================================================================
====
Learning rate tuning: lr = 0.001
==============================================================================
====
Epoch 01/3 | train_loss=0.191767 | val_loss=0.147525
Epoch 02/3 | train_loss=0.131763 | val_loss=0.130779
Epoch 03/3 | train_loss=0.120444 | val_loss=0.128468
==============================================================================
====
Learning rate tuning: lr = 0.0001
==============================================================================
====
Epoch 01/3 | train_loss=0.494228 | val_loss=0.365530
Epoch 02/3 | train_loss=0.252919 | val_loss=0.241398
Epoch 03/3 | train_loss=0.185322 | val_loss=0.196691
```

	learning_rate	best_val_loss	best_micro_f1	best_macro_f1	best_threshold_micro	b
0	0.0010	0.128468	0.120304	0.004833	0.65	
1	0.0001	0.196691	0.096637	0.002197	0.55	
2	0.0100	0.143856	0.075639	0.003817	0.55	

In [18]:
```python
best_lr_row = df_lr.iloc[0]
best_learning_rate = float(best_lr_row["learning_rate"])

print("Best learning rate:", best_learning_rate)
print("Best LR row:")
display(best_lr_row.to_frame())
```

```
Best learning rate: 0.001
Best LR row:
```

	0
learning_rate	0.001
best_val_loss	0.128468
best_micro_f1	0.120304
best_macro_f1	0.004833
best_threshold_micro	0.65
best_threshold_macro	0.55
model_path	kmer_lr_0.001.pt

Step 4. Regularization Strategy 1: Dropout

Why this step?

Dropout is a standard regularization technique that reduces overfitting by randomly masking hidden units during training.

What is fixed in this section?

We fix the best learning rate from the previous step and vary:

- `dropout`

Why this is a regularization experiment

Unlike learning rate, dropout does not primarily control optimization speed. Instead, it controls model complexity and encourages better generalization.

In [19]:
```
dropout_values = [0.0, 0.3, 0.5]
dropout_results = []

for dp in dropout_values:
    print("=" * 80)
    print(f"Dropout tuning: dropout = {dp}")
    print("=" * 80)

    config = TUNING_BASE_CONFIG.copy()
    config["learning_rate"] = best_learning_rate
    config["dropout"] = dp
    config["save_prefix"] = f"kmer_dropout_{dp}"

    result = run_experiment(
        config=config,
        train_dataset=train_dataset_small,
        val_dataset=val_dataset_small,
        input_dim=input_dim,
        output_dim=output_dim,
        criterion=criterion,
        device=device,
        run_threshold_sweep=True
    )

    dropout_results.append({
        "dropout": dp,
        "learning_rate": best_learning_rate,
        "best_val_loss": result["best_val_loss"],
        "best_micro_f1": result["best_micro_row"]["micro_f1"],
        "best_macro_f1": result["best_macro_row"]["macro_f1"],
        "best_threshold_micro": result["best_micro_row"]["threshold"],
        "model_path": result["model_path"]
    })
```

```
df_dropout = pd.DataFrame(dropout_results).sort_values("best_micro_f1", asce
display(df_dropout)
```

```
==============================================================================
====
Dropout tuning: dropout = 0.0
==============================================================================
====
Epoch 01/3 | train_loss=0.187249 | val_loss=0.138472
Epoch 02/3 | train_loss=0.122765 | val_loss=0.128738
Epoch 03/3 | train_loss=0.108448 | val_loss=0.123687
==============================================================================
====
Dropout tuning: dropout = 0.3
==============================================================================
====
Epoch 01/3 | train_loss=0.192109 | val_loss=0.144682
Epoch 02/3 | train_loss=0.131652 | val_loss=0.135308
Epoch 03/3 | train_loss=0.120453 | val_loss=0.128778
==============================================================================
====
Dropout tuning: dropout = 0.5
==============================================================================
====
Epoch 01/3 | train_loss=0.198240 | val_loss=0.163706
Epoch 02/3 | train_loss=0.139146 | val_loss=0.141415
Epoch 03/3 | train_loss=0.130260 | val_loss=0.133398
```

	dropout	learning_rate	best_val_loss	best_micro_f1	best_macro_f1	best_threshold
0	0.3	0.001	0.128778	0.118925	0.004672	
1	0.0	0.001	0.123687	0.117371	0.005665	
2	0.5	0.001	0.133398	0.104496	0.004046	

In [20]:
```
best_dropout_row = df_dropout.iloc[0]
best_dropout = float(best_dropout_row["dropout"])

print("Best dropout:", best_dropout)
print("Best dropout row:")
display(best_dropout_row.to_frame())
```

```
Best dropout: 0.3
Best dropout row:
```

	0
dropout	0.3
learning_rate	0.001
best_val_loss	0.128778
best_micro_f1	0.118925
best_macro_f1	0.004672
best_threshold_micro	0.65
model_path	kmer_dropout_0.3.pt

Step 5. Regularization Strategy 2: Weight Decay

Why this step?

Weight decay implements L2 regularization, which discourages overly large weights and can improve generalization.

What is fixed in this section?

We fix:

- the best learning rate
- the best dropout setting

We then vary:

- `weight_decay`

Why this section matters

This allows us to test whether explicit L2 regularization improves validation performance beyond dropout alone.

In [21]:
```
weight_decay_values = [0.0, 1e-5, 1e-4]
wd_results = []

for wd in weight_decay_values:
    print("=" * 80)
    print(f"Weight decay tuning: wd = {wd}")
    print("=" * 80)

    config = TUNING_BASE_CONFIG.copy()
    config["learning_rate"] = best_learning_rate
    config["dropout"] = best_dropout
    config["weight_decay"] = wd
```

```
    config["save_prefix"] = f"kmer_wd_{wd}"

    result = run_experiment(
        config=config,
        train_dataset=train_dataset_small,
        val_dataset=val_dataset_small,
        input_dim=input_dim,
        output_dim=output_dim,
        criterion=criterion,
        device=device,
        run_threshold_sweep=True
    )

    wd_results.append({
        "weight_decay": wd,
        "learning_rate": best_learning_rate,
        "dropout": best_dropout,
        "best_val_loss": result["best_val_loss"],
        "best_micro_f1": result["best_micro_row"]["micro_f1"],
        "best_macro_f1": result["best_macro_row"]["macro_f1"],
        "best_threshold_micro": result["best_micro_row"]["threshold"],
        "model_path": result["model_path"]
    })

df_wd = pd.DataFrame(wd_results).sort_values("best_micro_f1", ascending=Fals
display(df_wd)
```

```
================================================================================
====
Weight decay tuning: wd = 0.0
================================================================================
====
Epoch 01/3 | train_loss=0.188489 | val_loss=0.133086
Epoch 02/3 | train_loss=0.123393 | val_loss=0.126246
Epoch 03/3 | train_loss=0.111045 | val_loss=0.122710
================================================================================
====
Weight decay tuning: wd = 1e-05
================================================================================
====
Epoch 01/3 | train_loss=0.192248 | val_loss=0.141415
Epoch 02/3 | train_loss=0.131987 | val_loss=0.134198
Epoch 03/3 | train_loss=0.120237 | val_loss=0.128815
================================================================================
====
Weight decay tuning: wd = 0.0001
================================================================================
====
Epoch 01/3 | train_loss=0.269438 | val_loss=0.211412
Epoch 02/3 | train_loss=0.187120 | val_loss=0.187435
Epoch 03/3 | train_loss=0.169588 | val_loss=0.166680
```

	weight_decay	learning_rate	dropout	best_val_loss	best_micro_f1	best_macro_f1
0	0.00000	0.001	0.3	0.122710	0.124372	0.004561
1	0.00001	0.001	0.3	0.128815	0.111210	0.004829
2	0.00010	0.001	0.3	0.166680	0.109039	0.002977

In [22]:
```
best_wd_row = df_wd.iloc[0]
best_weight_decay = float(best_wd_row["weight_decay"])

print("Best weight decay:", best_weight_decay)
print("Best weight decay row:")
display(best_wd_row.to_frame())
```

```
Best weight decay: 0.0
Best weight decay row:
```

	0
weight_decay	0.0
learning_rate	0.001
dropout	0.3
best_val_loss	0.12271
best_micro_f1	0.124372
best_macro_f1	0.004561
best_threshold_micro	0.65
model_path	kmer_wd_0.0.pt

Step 6. Hidden Layer Size Tuning

Why this step?

Hidden layer size controls model capacity.
A model that is too small may underfit, while a model that is too large may overfit.

What we test

We compare several hidden-layer configurations while keeping the previously selected learning rate, dropout, and weight decay fixed.

In [23]:
```
hidden_configs = [
    (256, 128),
    (512, 256),
    (1024, 512)
]
```

```
hidden_results = []

for h1, h2 in hidden_configs:
    print("=" * 80)
    print(f"Hidden layer tuning: hidden1 = {h1}, hidden2 = {h2}")
    print("=" * 80)

    config = TUNING_BASE_CONFIG.copy()
    config["learning_rate"] = best_learning_rate
    config["dropout"] = best_dropout
    config["weight_decay"] = best_weight_decay

    config["hidden1"] = h1
    config["hidden2"] = h2

    config["save_prefix"] = f"kmer_hidden_{h1}_{h2}"

    result = run_experiment(
        config=config,
        train_dataset=train_dataset_small,
        val_dataset=val_dataset_small,
        input_dim=input_dim,
        output_dim=output_dim,
        criterion=criterion,
        device=device,
        run_threshold_sweep=True
    )

    hidden_results.append({
        "hidden1": h1,
        "hidden2": h2,
        "learning_rate": best_learning_rate,
        "dropout": best_dropout,
        "weight_decay": best_weight_decay,
        "best_val_loss": result["best_val_loss"],
        "best_micro_f1": result["best_micro_row"]["micro_f1"],
        "best_macro_f1": result["best_macro_row"]["macro_f1"],
        "best_threshold_micro": result["best_micro_row"]["threshold"],
        "model_path": result["model_path"]
    })

df_hidden = pd.DataFrame(hidden_results)\
    .sort_values("best_micro_f1", ascending=False)\
    .reset_index(drop=True)

display(df_hidden)
```

```
==============================================================================
====
Hidden layer tuning: hidden1 = 256, hidden2 = 128
==============================================================================
====
Epoch 01/3 | train_loss=0.233393 | val_loss=0.142216
Epoch 02/3 | train_loss=0.130835 | val_loss=0.129332
Epoch 03/3 | train_loss=0.120152 | val_loss=0.125514
==============================================================================
====
Hidden layer tuning: hidden1 = 512, hidden2 = 256
==============================================================================
====
Epoch 01/3 | train_loss=0.187814 | val_loss=0.134163
Epoch 02/3 | train_loss=0.123682 | val_loss=0.126424
Epoch 03/3 | train_loss=0.110980 | val_loss=0.122869
==============================================================================
====
Hidden layer tuning: hidden1 = 1024, hidden2 = 512
==============================================================================
====
Epoch 01/3 | train_loss=0.167297 | val_loss=0.129602
Epoch 02/3 | train_loss=0.117415 | val_loss=0.123721
Epoch 03/3 | train_loss=0.101200 | val_loss=0.122067
```

	hidden1	hidden2	learning_rate	dropout	weight_decay	best_val_loss	best_micro_
0	1024	512	0.001	0.3	0.0	0.122067	0.12530
1	256	128	0.001	0.3	0.0	0.125514	0.11769
2	512	256	0.001	0.3	0.0	0.122869	0.11656

In [24]:

```
best_hidden_row = df_hidden.iloc[0]

best_hidden1 = int(best_hidden_row["hidden1"])
best_hidden2 = int(best_hidden_row["hidden2"])

print("Best hidden layer:")
print(f"hidden1 = {best_hidden1}, hidden2 = {best_hidden2}")

print("Best hidden row:")
display(best_hidden_row.to_frame())
```

```
Best hidden layer:
hidden1 = 1024, hidden2 = 512
Best hidden row:
```

	0
hidden1	1024
hidden2	512
learning_rate	0.001
dropout	0.3
weight_decay	0.0
best_val_loss	0.122067
best_micro_f1	0.125309
best_macro_f1	0.006195
best_threshold_micro	0.65
model_path	kmer_hidden_1024_512.pt

Step 6. Final Model with Best Hyperparameters

Why this step?

After tuning optimization and regularization settings separately, we now combine the best choices into one final configuration.

What is included in the final configuration?

- best learning rate
- best dropout
- best weight decay
- early stopping during training

Why threshold sweep is used here

This is the stage where we want final threshold-based evaluation, so we now run full validation threshold sweeping and record the best micro-F1 and macro-F1.

In [25]:
```
FINAL_CONFIG = TUNING_BASE_CONFIG.copy()
FINAL_CONFIG["learning_rate"] = best_learning_rate
FINAL_CONFIG["dropout"] = best_dropout
FINAL_CONFIG["weight_decay"] = best_weight_decay
FINAL_CONFIG["save_prefix"] = "kmer_final_best_model"
FINAL_CONFIG["hidden1"] = best_hidden1
FINAL_CONFIG["hidden2"] = best_hidden2
FINAL_CONFIG["num_epochs"] = 5

print("Final config:")
print(FINAL_CONFIG)
```

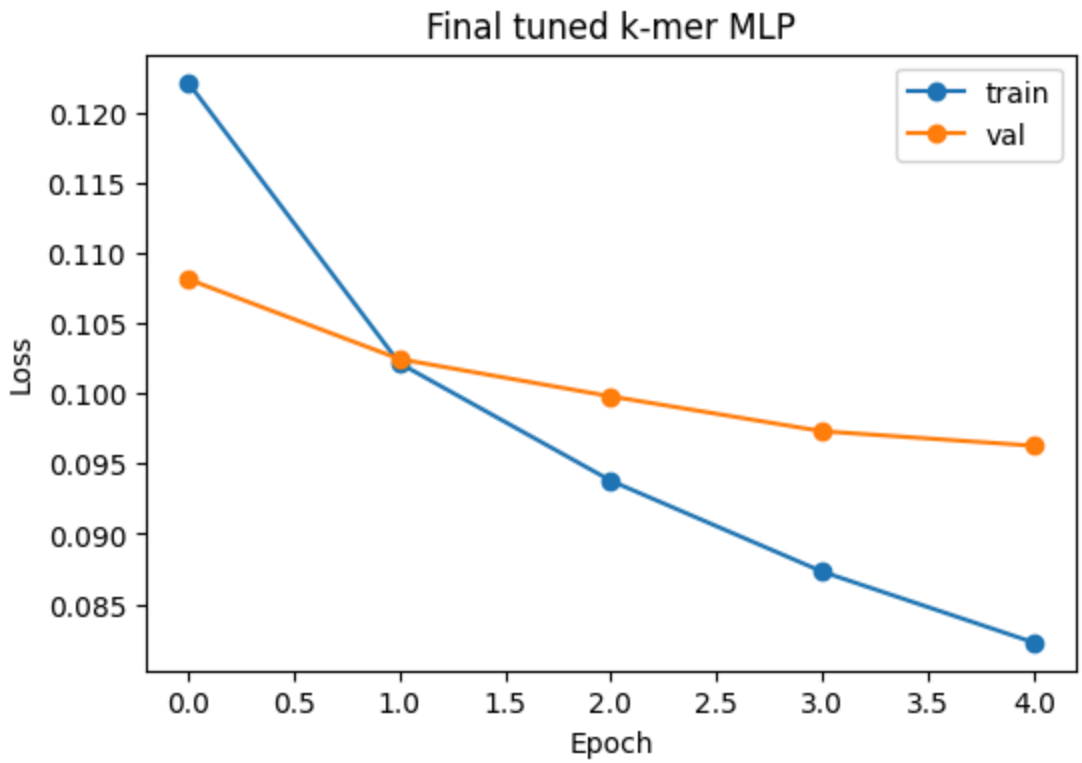
```
Final config:
{'hidden1': 1024, 'hidden2': 512, 'dropout': 0.3, 'learning_rate': 0.001, 'w
eight_decay': 0.0, 'batch_size': 64, 'num_epochs': 5, 'patience': 3, 'save_p
refix': 'kmer_final_best_model'}
```

In [26]:
```
final_result = run_experiment(
    config=FINAL_CONFIG,
    train_dataset=train_dataset,
    val_dataset=val_dataset,
    input_dim=input_dim,
    output_dim=output_dim,
    criterion=criterion,
    device=device,
    run_threshold_sweep=True
)

print("Final best val loss:", final_result["best_val_loss"])
print("Final best micro-F1:", final_result["best_micro_row"])
print("Final best macro-F1:", final_result["best_macro_row"])
print("Final model path:", final_result["model_path"])
```

```
Epoch 01/5 | train_loss=0.122105 | val_loss=0.108112
Epoch 02/5 | train_loss=0.102138 | val_loss=0.102431
Epoch 03/5 | train_loss=0.093778 | val_loss=0.099776
Epoch 04/5 | train_loss=0.087310 | val_loss=0.097298
Epoch 05/5 | train_loss=0.082274 | val_loss=0.096266
Final best val loss: 0.09626565076072108
Final best micro-F1: {'threshold': 0.7500000000000002, 'micro_f1': 0.1503630
8447686397, 'macro_f1': 0.042363616960929}
Final best macro-F1: {'threshold': 0.6500000000000001, 'micro_f1': 0.1480968
4198717687, 'macro_f1': 0.04641646766062745}
Final model path: kmer_final_best_model.pt
```

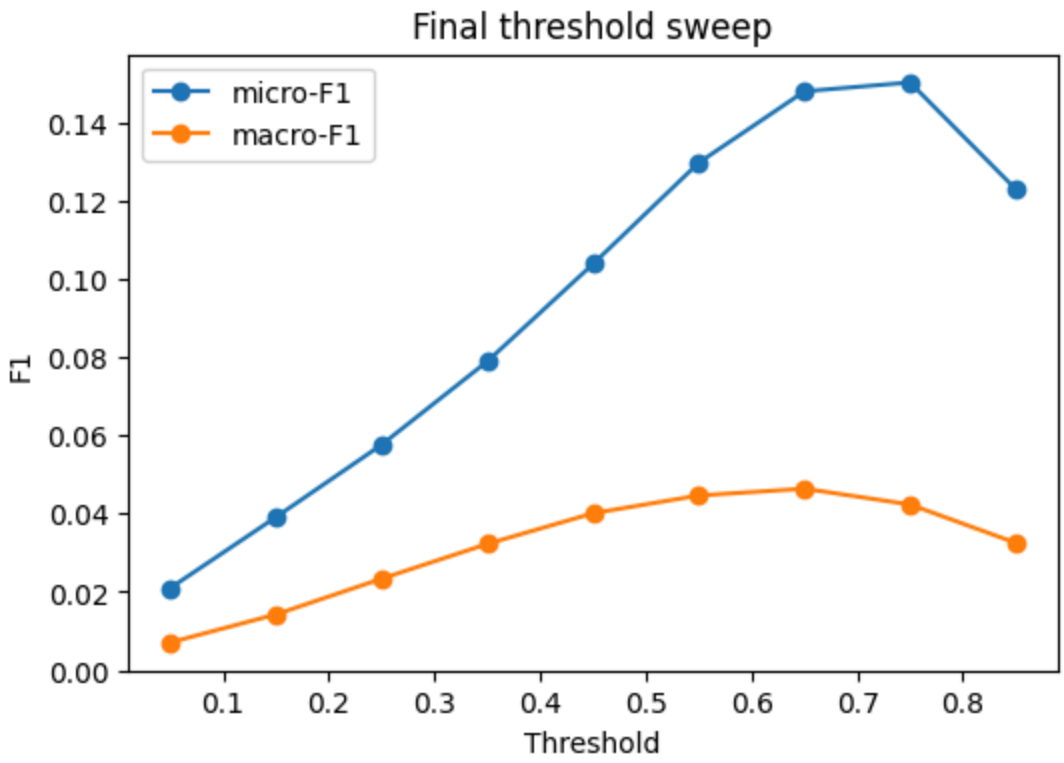
In [27]:
```
plt.figure(figsize=(6, 4))
plt.plot(final_result["train_losses"], marker="o", label="train")
plt.plot(final_result["val_losses"], marker="o", label="val")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Final tuned k-mer MLP")
plt.legend()
plt.show()
```



In [28]:
```
df_thresh = pd.DataFrame(final_result["threshold_records"])
display(df_thresh)

plt.figure(figsize=(6, 4))
plt.plot(df_thresh["threshold"], df_thresh["micro_f1"], marker="o", label="m
plt.plot(df_thresh["threshold"], df_thresh["macro_f1"], marker="o", label="m
plt.xlabel("Threshold")
plt.ylabel("F1")
plt.title("Final threshold sweep")
plt.legend()
plt.show()
```

	threshold	micro_f1	macro_f1
0	0.05	0.020996	0.007070
1	0.15	0.039147	0.014242
2	0.25	0.057668	0.023386
3	0.35	0.079003	0.032279
4	0.45	0.103917	0.040130
5	0.55	0.129820	0.044667
6	0.65	0.148097	0.046416
7	0.75	0.150363	0.042364
8	0.85	0.123033	0.032545



Step 7. Save the Selected Hyperparameters and Final Results

Why this step?

Saving the selected configuration makes the experiment reproducible and lets us reuse the tuned settings later, for example:

- as the final k-mer baseline
- as a starting point for future comparisons
- as a template for the ESM notebook

What we save

We save:

- the best learning rate
- the best dropout value
- the best weight decay value
- the final validation metrics
- the best threshold information

In [29]:
```
summary = {
    "best_learning_rate": best_learning_rate,
    "best_dropout": best_dropout,
    "best_weight_decay": best_weight_decay,
```

```
    "final_best_val_loss": float(final_result["best_val_loss"]),
    "final_best_micro": final_result["best_micro_row"],
    "final_best_macro": final_result["best_macro_row"],
    "final_model_path": final_result["model_path"],
}

with open("kmer_tuning_summary.json", "w") as f:
    json.dump(summary, f, indent=2)

print("Saved tuning summary to kmer_tuning_summary.json")
print(summary)
```

```
Saved tuning summary to kmer_tuning_summary.json
{'best_learning_rate': 0.001, 'best_dropout': 0.3, 'best_weight_decay': 0.0,
'final_best_val_loss': 0.09626565076072108, 'final_best_micro': {'threshol
d': 0.7500000000000002, 'micro_f1': 0.15036308447686397, 'macro_f1': 0.04236
3616960929}, 'final_best_macro': {'threshold': 0.6500000000000001, 'micro_f
1': 0.14809684198717687, 'macro_f1': 0.04641646766062745}, 'final_model_pat
h': 'kmer_final_best_model.pt'}
```

Summary and Interpretation

What this notebook accomplished

This notebook reorganized the k-mer MLP workflow so that the early sections only perform a short baseline sanity check, while the main emphasis is placed on:

- hyperparameter tuning
- regularization strategies
- final threshold-based model selection

Key methodological choices

- short baseline run without threshold sweeping
- learning rate tuned first as an optimization hyperparameter
- dropout and weight decay evaluated as regularization strategies
- early stopping used during all tuning runs
- final threshold sweep used only after selecting the best settings

Why this structure is appropriate

This notebook reflects the intended focus of the week's assignment:

- identifying stable training settings
- reducing overfitting
- improving validation generalization
- documenting the selected final hyperparameters

ESM MLP: Baseline, Regularization, and Hyperparameter Tuning

Notebook goal

This notebook starts from the **ESM workflow** and keeps the same train/validation split strategy used in `MLP_ESM.ipynb`.
It then adopts the **small-subset tuning strategy** discussed for faster hyperparameter search.

What this notebook does

1. Load precomputed ESM embeddings and aligned GO labels
2. Reuse the same `train_idx` / `val_idx` split from the k-mer pipeline
3. Standardize the ESM features using training-only statistics
4. Build the ESM dataset and shared training utilities
5. Create small subsets for fast tuning
6. Tune:
   - learning rate
   - dropout
   - weight decay
   - hidden layer size
7. Train a final ESM model on the full training/validation split using the selected hyperparameters
8. Perform final threshold sweeping and save a summary

Why use small subsets first?

Full GO prediction is computationally expensive.
Small-subset tuning allows us to identify promising hyperparameter ranges more efficiently before running a final full-data model.

Step 0. Imports and ESM Data Loading

Why this step?

We first load the precomputed ESM embeddings and the aligned GO label matrix.
This notebook assumes that the feature extraction stage has already been completed.

What we load

- `X_esm.npy` : mean-pooled ESM embeddings
- `Y_sparse.npz` or `Y.npy` : GO target matrix
- `protein_ids.pkl` : aligned protein IDs, if available

Why this matters

Separating feature extraction from supervised learning makes the tuning stage much faster and easier to reproduce.

In [1]:
```
import os
import json
import copy
import pickle
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from scipy.sparse import load_npz

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader, Subset

from sklearn.preprocessing import StandardScaler
from sklearn.metrics import f1_score

DATA_DIR = "data_processed_esm"

X_esm = np.load(os.path.join(DATA_DIR, "X_esm.npy"))
print("X_esm shape:", X_esm.shape, X_esm.dtype)

y_sparse_path = os.path.join(DATA_DIR, "Y_sparse.npz")
y_dense_path = os.path.join(DATA_DIR, "Y.npy")

if os.path.exists(y_sparse_path):
    Y = load_npz(y_sparse_path)
    print("Loaded sparse Y:", Y.shape, type(Y))
elif os.path.exists(y_dense_path):
    Y = np.load(y_dense_path)
    print("Loaded dense Y:", Y.shape, type(Y))
else:
    raise FileNotFoundError("Cannot find Y_sparse.npz or Y.npy")

protein_ids_path = os.path.join(DATA_DIR, "protein_ids.pkl")
if os.path.exists(protein_ids_path):
    with open(protein_ids_path, "rb") as f:
        protein_ids = pickle.load(f)
    print("protein_ids length:", len(protein_ids))
```

```
X_esm shape: (140569, 640) float32
Loaded sparse Y: (140569, 31454) <class 'scipy.sparse._csr.csr_matrix'>
```

Step 1. Reuse the Same Train/Validation Split as `MLP_ESM`

Why this step?

To compare models fairly, we reuse the same train/validation split strategy as the original ESM notebook:

- read `train_idx.npy`
- read `val_idx.npy`
- apply them to the aligned ESM feature matrix and GO labels

Why this matters

This ensures that:

- the ESM tuning experiments remain consistent with the original workflow
- comparisons across models are made on the same protein samples

In [2]:
```
print("Number of ESM samples:", X_esm.shape[0])
print("Number of Y samples:", Y.shape[0])
assert X_esm.shape[0] == Y.shape[0], "X_esm and Y are not aligned!"

train_idx = np.load("data_processed_kmer/train_idx.npy")
val_idx   = np.load("data_processed_kmer/val_idx.npy")

X_train = X_esm[train_idx]
X_val   = X_esm[val_idx]

if hasattr(Y, "tocsr"):
    Y_train = Y[train_idx]
    Y_val   = Y[val_idx]
else:
    Y_train = Y[train_idx]
    Y_val   = Y[val_idx]

print("X_train:", X_train.shape)
print("X_val:  ", X_val.shape)
print("Y_train:", Y_train.shape)
print("Y_val:  ", Y_val.shape)
```

```
Number of ESM samples: 140569
Number of Y samples: 140569
X_train: (112455, 640)
X_val:   (28114, 640)
Y_train: (112455, 31454)
Y_val:   (28114, 31454)
```

Step 2. Standardize the ESM Features

Why this step?

Even though ESM embeddings are pretrained features, standardization can still improve downstream optimization stability.

Important design choice

The scaler is fit **only on the training set**, then applied to the validation set. This prevents data leakage.

In [3]:
```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train).astype(np.float32)
X_val   = scaler.transform(X_val).astype(np.float32)

with open("esm_scaler.pkl", "wb") as f:
    pickle.dump(scaler, f)

print("Standardized X_train:", X_train.shape, X_train.dtype)
print("Standardized X_val:  ", X_val.shape, X_val.dtype)
```

```
Standardized X_train: (112455, 640) float32
Standardized X_val:   (28114, 640) float32
```

Step 3. Define the ESM Dataset and Shared Training Components

Why this step?

All tuning experiments will reuse the same:

- dataset class
- model class
- training loop
- evaluation loop
- threshold sweep
- experiment wrapper

Why organize it this way?

This keeps the earlier data preparation fixed while letting us vary hyperparameters cleanly in later sections.

In [4]:
```
class ProteinESMDataset(Dataset):
    def __init__(self, X, Y):
        self.X = X.astype(np.float32)
```

```
        self.Y = Y
        self.is_sparse = hasattr(Y, "tocsr")
        if self.is_sparse:
            self.Y = Y.tocsr()
            self.n_labels = self.Y.shape[1]

    def __len__(self):
        return self.X.shape[0]

    def __getitem__(self, idx):
        x = torch.from_numpy(self.X[idx])

        if self.is_sparse:
            start = self.Y.indptr[idx]
            end = self.Y.indptr[idx + 1]
            pos_idx = self.Y.indices[start:end]

            y = torch.zeros(self.n_labels, dtype=torch.float32)
            y[pos_idx] = 1.0
        else:
            y = torch.tensor(self.Y[idx], dtype=torch.float32)

        return x, y


train_dataset = ProteinESMDataset(X_train, Y_train)
val_dataset   = ProteinESMDataset(X_val, Y_val)

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
input_dim = X_train.shape[1]
output_dim = Y_train.shape[1]

print("Device:", device)
print("Input dim:", input_dim)
print("Output dim:", output_dim)
```

```
Device: cpu
Input dim: 640
Output dim: 31454
```

```
/global/home/users/yushanfu/esm_env/lib/python3.11/site-packages/torch/cuda/
__init__.py:180: UserWarning: CUDA initialization: The NVIDIA driver on your
system is too old (found version 12030). Please update your GPU driver by do
wnloading and installing a new version from the URL: http://www.nvidia.com/D
ownload/index.aspx Alternatively, go to: https://pytorch.org to install a Py
Torch version that has been compiled with your version of the CUDA driver.
(Triggered internally at /pytorch/c10/cuda/CUDAFunctions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
```

In [5]:

```
class ESMMLP(nn.Module):
    def __init__(self, input_dim, output_dim, hidden1=512, hidden2=256, drop
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden1),
            nn.BatchNorm1d(hidden1),
            nn.ReLU(),
            nn.Dropout(dropout),
```

```
            nn.Linear(hidden1, hidden2),
            nn.BatchNorm1d(hidden2),
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden2, output_dim)
        )

    def forward(self, x):
        return self.net(x)


if hasattr(Y_train, "sum") and not isinstance(Y_train, np.ndarray):
    label_counts = np.asarray(Y_train.sum(axis=0)).ravel().astype(np.float32
else:
    label_counts = Y_train.sum(axis=0).astype(np.float32)

n_train = Y_train.shape[0]
pos_weight = (n_train - label_counts) / (label_counts + 1e-6)
pos_weight = np.clip(pos_weight, 1.0, 100.0)

print("label_counts shape:", label_counts.shape)
print("pos_weight min/max:", pos_weight.min(), pos_weight.max())

pos_weight_tensor = torch.tensor(pos_weight, dtype=torch.float32, device=dev
criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight_tensor)
```

```
label_counts shape: (31454,)
pos_weight min/max: 1.0 100.0
```

In [6]:

```
def build_dataloaders(train_dataset, val_dataset, batch_size):
    train_loader = DataLoader(
        train_dataset,
        batch_size=batch_size,
        shuffle=True,
        num_workers=4,
        pin_memory=torch.cuda.is_available()
    )

    val_loader = DataLoader(
        val_dataset,
        batch_size=batch_size,
        shuffle=False,
        num_workers=4,
        pin_memory=torch.cuda.is_available()
    )
    return train_loader, val_loader


def build_model(config, input_dim, output_dim, device):
    return ESMMLP(
        input_dim=input_dim,
        output_dim=output_dim,
        hidden1=config["hidden1"],
        hidden2=config["hidden2"],
```

```
        dropout=config["dropout"]
    ).to(device)


def build_optimizer(model, config):
    return optim.Adam(
        model.parameters(),
        lr=config["learning_rate"],
        weight_decay=config["weight_decay"]
    )


def train_one_epoch(model, loader, criterion, optimizer, device):
    model.train()
    total_loss = 0.0

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        Y_batch = Y_batch.to(device, non_blocking=True)

        optimizer.zero_grad()
        logits = model(X_batch)
        loss = criterion(logits, Y_batch)
        loss.backward()
        optimizer.step()

        total_loss += loss.item()

    return total_loss / len(loader)


@torch.no_grad()
def evaluate_loss(model, loader, criterion, device):
    model.eval()
    total_loss = 0.0

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        Y_batch = Y_batch.to(device, non_blocking=True)

        logits = model(X_batch)
        loss = criterion(logits, Y_batch)

        total_loss += loss.item()

    return total_loss / len(loader)


@torch.no_grad()
def get_predictions(model, loader, device):
    model.eval()
    all_probs = []
    all_targets = []

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
```

```
        logits = model(X_batch)
        probs = torch.sigmoid(logits).cpu().numpy()

        all_probs.append(probs)
        all_targets.append(Y_batch.numpy())

    all_probs = np.vstack(all_probs)
    all_targets = np.vstack(all_targets)

    return all_probs, all_targets


def threshold_sweep(y_true, y_prob, thresholds=None):
    if thresholds is None:
        thresholds = np.arange(0.05, 0.95, 0.1)

    records = []
    for t in thresholds:
        y_pred = (y_prob >= t).astype(np.int8)

        micro = f1_score(y_true, y_pred, average="micro", zero_division=0)
        macro = f1_score(y_true, y_pred, average="macro", zero_division=0)

        records.append({
            "threshold": float(t),
            "micro_f1": float(micro),
            "macro_f1": float(macro)
        })

    best_micro_row = max(records, key=lambda x: x["micro_f1"])
    best_macro_row = max(records, key=lambda x: x["macro_f1"])
    return records, best_micro_row, best_macro_row


def train_with_early_stopping(model, train_loader, val_loader, criterion, op
                              num_epochs=10, patience=3, save_path=None):
    best_val_loss = float("inf")
    best_state_dict = None
    epochs_no_improve = 0

    train_losses = []
    val_losses = []

    for epoch in range(num_epochs):
        train_loss = train_one_epoch(model, train_loader, criterion, optimiz
        val_loss = evaluate_loss(model, val_loader, criterion, device)

        train_losses.append(train_loss)
        val_losses.append(val_loss)

        print(f"Epoch {epoch+1:02d}/{num_epochs} | train_loss={train_loss:.6

        if val_loss < best_val_loss:
            best_val_loss = val_loss
            best_state_dict = copy.deepcopy(model.state_dict())
```

```
            epochs_no_improve = 0

            if save_path is not None:
                torch.save(best_state_dict, save_path)
        else:
            epochs_no_improve += 1

        if epochs_no_improve >= patience:
            print(f"Early stopping triggered at epoch {epoch+1}")
            break

    if best_state_dict is not None:
        model.load_state_dict(best_state_dict)

    return model, train_losses, val_losses, best_val_loss


def run_experiment(config, train_dataset, val_dataset, input_dim, output_dim
                   run_threshold_sweep=True):
    train_loader, val_loader = build_dataloaders(
        train_dataset=train_dataset,
        val_dataset=val_dataset,
        batch_size=config["batch_size"]
    )

    model = build_model(config, input_dim, output_dim, device)
    optimizer = build_optimizer(model, config)
    save_path = f"{config['save_prefix']}.pt"

    model, train_losses, val_losses, best_val_loss = train_with_early_stoppi
        model=model,
        train_loader=train_loader,
        val_loader=val_loader,
        criterion=criterion,
        optimizer=optimizer,
        device=device,
        num_epochs=config["num_epochs"],
        patience=config["patience"],
        save_path=save_path
    )

    result = {
        "config": config.copy(),
        "model": model,
        "model_path": save_path,
        "best_val_loss": best_val_loss,
        "train_losses": train_losses,
        "val_losses": val_losses,
    }

    if run_threshold_sweep:
        val_probs, val_targets = get_predictions(model, val_loader, device)
        records, best_micro_row, best_macro_row = threshold_sweep(val_target
        result["threshold_records"] = records
        result["best_micro_row"] = best_micro_row
        result["best_macro_row"] = best_macro_row
```

```
    return result
```

Step 4. Build Small Subsets for Fast Tuning

Why this step?

Full-data tuning is expensive.
We therefore create smaller subsets of the training and validation data to run faster coarse experiments.

Important note

These subset experiments are used to identify promising hyperparameter settings, not to produce the final reported model.

In [7]:
```
np.random.seed(42)

small_train_size = len(train_dataset) // 10
small_val_size = len(val_dataset) // 10

small_train_idx = np.random.choice(len(train_dataset), size=small_train_size
small_val_idx = np.random.choice(len(val_dataset), size=small_val_size, repl

train_dataset_small = Subset(train_dataset, small_train_idx)
val_dataset_small = Subset(val_dataset, small_val_idx)

print("Full train size:", len(train_dataset))
print("Full val size:", len(val_dataset))
print("Small train size:", len(train_dataset_small))
print("Small val size:", len(val_dataset_small))
```

```
Full train size: 112455
Full val size: 28114
Small train size: 11245
Small val size: 2811
```

Step 5. Quick Baseline Sanity Check on Small Subsets

Why this step?

Before running full tuning loops, we perform a short sanity check to verify that:

- the ESM pipeline runs correctly
- the loss stays finite
- checkpointing and dataloaders behave as expected

Important note

We keep this short and do **not** use it as the final reported model.

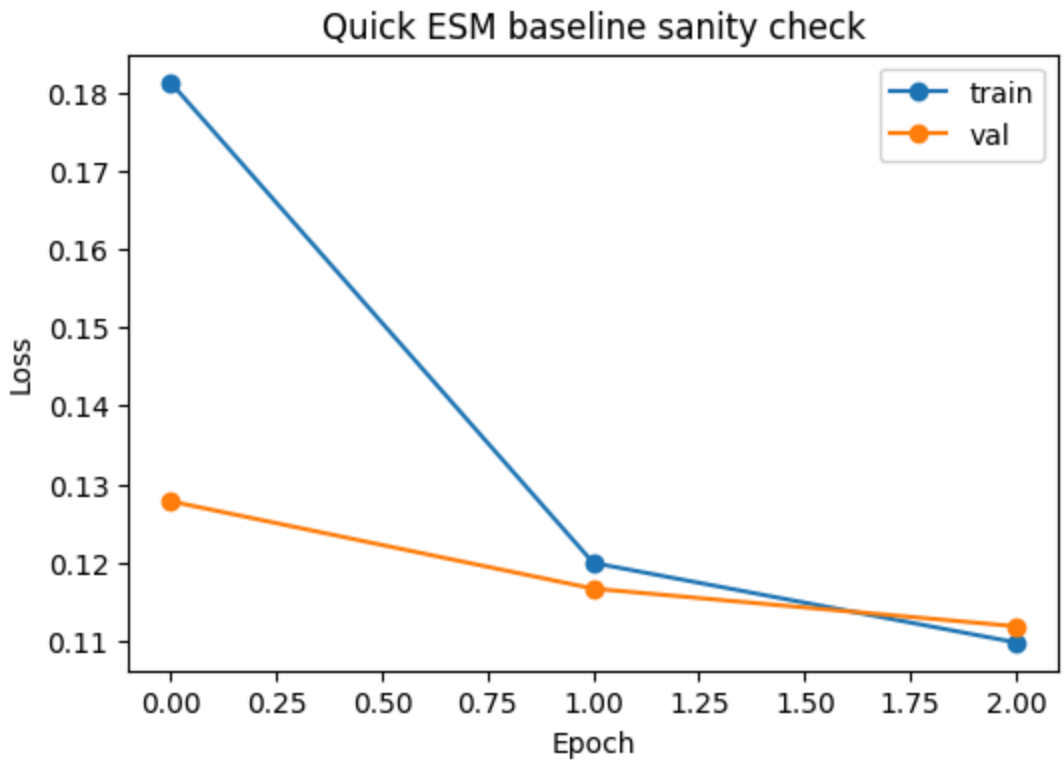
In [8]:
```
ESM_BASELINE_CONFIG = {
    "hidden1": 512,
    "hidden2": 256,
    "dropout": 0.3,
    "learning_rate": 1e-3,
    "weight_decay": 1e-5,
    "batch_size": 64,
    "num_epochs": 3,
    "patience": 2,
    "save_prefix": "esm_baseline_small"
}

baseline_result_esm = run_experiment(
    config=ESM_BASELINE_CONFIG,
    train_dataset=train_dataset_small,
    val_dataset=val_dataset_small,
    input_dim=input_dim,
    output_dim=output_dim,
    criterion=criterion,
    device=device,
    run_threshold_sweep=False
)

print("ESM baseline best val loss:", baseline_result_esm["best_val_loss"])
print("ESM baseline model path:", baseline_result_esm["model_path"])
```

```
Epoch 01/3 | train_loss=0.181280 | val_loss=0.127801
Epoch 02/3 | train_loss=0.119930 | val_loss=0.116648
Epoch 03/3 | train_loss=0.109776 | val_loss=0.111840
ESM baseline best val loss: 0.11183971539139748
ESM baseline model path: esm_baseline_small.pt
```

In [9]:
```
plt.figure(figsize=(6, 4))
plt.plot(baseline_result_esm["train_losses"], marker="o", label="train")
plt.plot(baseline_result_esm["val_losses"], marker="o", label="val")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Quick ESM baseline sanity check")
plt.legend()
plt.show()
```



Step 6. Hyperparameter Tuning: Learning Rate

Why this step?

We tune the learning rate first because it strongly affects optimization stability and convergence speed.

What is fixed in this section?

We keep the architecture and regularization settings fixed, and only vary:

- `learning_rate`

Selection rule

For each run, we perform threshold sweeping on the validation subset and select the best learning rate using **best micro-F1**.

In [10]:
```
ESM_TUNING_BASE_CONFIG = {
    "hidden1": 512,
    "hidden2": 256,
    "dropout": 0.3,
    "learning_rate": 1e-3,
    "weight_decay": 1e-5,
    "batch_size": 64,
    "num_epochs": 3,
    "patience": 3,
```

```
    "save_prefix": "esm_tuning_run"
}

learning_rates = [1e-2, 1e-3, 1e-4]
lr_results_esm = []

for lr in learning_rates:
    print("=" * 80)
    print(f"ESM learning rate tuning: lr = {lr}")
    print("=" * 80)

    config = ESM_TUNING_BASE_CONFIG.copy()
    config["learning_rate"] = lr
    config["save_prefix"] = f"esm_lr_{lr}"

    result = run_experiment(
        config=config,
        train_dataset=train_dataset_small,
        val_dataset=val_dataset_small,
        input_dim=input_dim,
        output_dim=output_dim,
        criterion=criterion,
        device=device,
        run_threshold_sweep=True
    )

    lr_results_esm.append({
        "learning_rate": lr,
        "best_val_loss": result["best_val_loss"],
        "best_micro_f1": result["best_micro_row"]["micro_f1"],
        "best_macro_f1": result["best_micro_row"]["macro_f1"],
        "best_threshold_micro": result["best_micro_row"]["threshold"],
        "model_path": result["model_path"]
    })

df_lr_esm = pd.DataFrame(lr_results_esm).sort_values("best_micro_f1", ascenc
display(df_lr_esm)
```

```
==============================================================================
====
ESM learning rate tuning: lr = 0.01
==============================================================================
====
Epoch 01/3 | train_loss=0.156106 | val_loss=0.138144
Epoch 02/3 | train_loss=0.138327 | val_loss=0.134944
Epoch 03/3 | train_loss=0.135681 | val_loss=0.133262
==============================================================================
====
ESM learning rate tuning: lr = 0.001
==============================================================================
====
Epoch 01/3 | train_loss=0.181230 | val_loss=0.129160
Epoch 02/3 | train_loss=0.120129 | val_loss=0.117521
Epoch 03/3 | train_loss=0.109869 | val_loss=0.111661
==============================================================================
====
ESM learning rate tuning: lr = 0.0001
==============================================================================
====
Epoch 01/3 | train_loss=0.484163 | val_loss=0.342315
Epoch 02/3 | train_loss=0.239764 | val_loss=0.213064
Epoch 03/3 | train_loss=0.173557 | val_loss=0.169911
```

	learning_rate	best_val_loss	best_micro_f1	best_macro_f1	best_threshold_micro	
0	0.0010	0.111661	0.156353	0.007186	0.75	
1	0.0001	0.169911	0.139197	0.002980	0.55	e
2	0.0100	0.133262	0.083634	0.005785	0.65	

In [11]:
```python
best_lr_row_esm = df_lr_esm.iloc[0]
best_learning_rate_esm = float(best_lr_row_esm["learning_rate"])

print("Best ESM learning rate:", best_learning_rate_esm)
print("Best ESM LR row:")
display(best_lr_row_esm.to_frame())
```

```
Best ESM learning rate: 0.001
Best ESM LR row:
```

	0
learning_rate	0.001
best_val_loss	0.111661
best_micro_f1	0.156353
best_macro_f1	0.007186
best_threshold_micro	0.75
model_path	esm_lr_0.001.pt

Step 7. Regularization Strategy 1: Dropout

Why this step?

Dropout is a standard regularization technique that reduces overfitting by randomly masking hidden units during training.

What is fixed in this section?

We fix the best learning rate from the previous step and vary:

- `dropout`

In [12]:
```
dropout_values = [0.0, 0.3, 0.5]
dropout_results_esm = []

for dp in dropout_values:
    print("=" * 80)
    print(f"ESM dropout tuning: dropout = {dp}")
    print("=" * 80)

    config = ESM_TUNING_BASE_CONFIG.copy()
    config["learning_rate"] = best_learning_rate_esm
    config["dropout"] = dp
    config["save_prefix"] = f"esm_dropout_{dp}"

    result = run_experiment(
        config=config,
        train_dataset=train_dataset_small,
        val_dataset=val_dataset_small,
        input_dim=input_dim,
        output_dim=output_dim,
        criterion=criterion,
        device=device,
        run_threshold_sweep=True
    )

    dropout_results_esm.append({
        "dropout": dp,
        "learning_rate": best_learning_rate_esm,
        "best_val_loss": result["best_val_loss"],
        "best_micro_f1": result["best_micro_row"]["micro_f1"],
        "best_macro_f1": result["best_micro_row"]["macro_f1"],
        "best_threshold_micro": result["best_micro_row"]["threshold"],
        "model_path": result["model_path"]
    })

df_dropout_esm = pd.DataFrame(dropout_results_esm).sort_values("best_micro_f
display(df_dropout_esm)
```

```
==============================================================================
====
ESM dropout tuning: dropout = 0.0
==============================================================================
====
Epoch 01/3 | train_loss=0.173562 | val_loss=0.123479
Epoch 02/3 | train_loss=0.113475 | val_loss=0.115382
Epoch 03/3 | train_loss=0.102288 | val_loss=0.108750
==============================================================================
====
ESM dropout tuning: dropout = 0.3
==============================================================================
====
Epoch 01/3 | train_loss=0.181455 | val_loss=0.126710
Epoch 02/3 | train_loss=0.120110 | val_loss=0.116311
Epoch 03/3 | train_loss=0.109736 | val_loss=0.112144
==============================================================================
====
ESM dropout tuning: dropout = 0.5
==============================================================================
====
Epoch 01/3 | train_loss=0.190585 | val_loss=0.134550
Epoch 02/3 | train_loss=0.127949 | val_loss=0.122496
Epoch 03/3 | train_loss=0.117981 | val_loss=0.116111
```

	dropout	learning_rate	best_val_loss	best_micro_f1	best_macro_f1	best_threshold
0	0.3	0.001	0.112144	0.151040	0.007141	
1	0.0	0.001	0.108750	0.149645	0.008413	
2	0.5	0.001	0.116111	0.134258	0.005917	

In [13]:
```
best_dropout_row_esm = df_dropout_esm.iloc[0]
best_dropout_esm = float(best_dropout_row_esm["dropout"])

print("Best ESM dropout:", best_dropout_esm)
print("Best ESM dropout row:")
display(best_dropout_row_esm.to_frame())
```

```
Best ESM dropout: 0.3
Best ESM dropout row:
```

	0
dropout	0.3
learning_rate	0.001
best_val_loss	0.112144
best_micro_f1	0.15104
best_macro_f1	0.007141
best_threshold_micro	0.65
model_path	esm_dropout_0.3.pt

Step 8. Regularization Strategy 2: Weight Decay

Why this step?

Weight decay implements L2 regularization, which discourages overly large weights and can improve generalization.

What is fixed in this section?

We fix:

- the best learning rate
- the best dropout setting

We then vary:

- `weight_decay`

In [14]:
```
weight_decay_values = [0.0, 1e-5, 1e-4]
wd_results_esm = []

for wd in weight_decay_values:
    print("=" * 80)
    print(f"ESM weight decay tuning: wd = {wd}")
    print("=" * 80)

    config = ESM_TUNING_BASE_CONFIG.copy()
    config["learning_rate"] = best_learning_rate_esm
    config["dropout"] = best_dropout_esm
    config["weight_decay"] = wd
    config["save_prefix"] = f"esm_wd_{wd}"

    result = run_experiment(
        config=config,
        train_dataset=train_dataset_small,
        val_dataset=val_dataset_small,
        input_dim=input_dim,
        output_dim=output_dim,
        criterion=criterion,
        device=device,
        run_threshold_sweep=True
    )

    wd_results_esm.append({
        "weight_decay": wd,
        "learning_rate": best_learning_rate_esm,
        "dropout": best_dropout_esm,
        "best_val_loss": result["best_val_loss"],
        "best_micro_f1": result["best_micro_row"]["micro_f1"],
        "best_macro_f1": result["best_micro_row"]["macro_f1"],
        "best_threshold_micro": result["best_micro_row"]["threshold"],
```

```
        "model_path": result["model_path"]
    })

df_wd_esm = pd.DataFrame(wd_results_esm).sort_values("best_micro_f1", ascenc
display(df_wd_esm)
```

```
================================================================================
====
ESM weight decay tuning: wd = 0.0
================================================================================
====
Epoch 01/3 | train_loss=0.177957 | val_loss=0.119755
Epoch 02/3 | train_loss=0.111438 | val_loss=0.110954
Epoch 03/3 | train_loss=0.099386 | val_loss=0.106917
================================================================================
====
ESM weight decay tuning: wd = 1e-05
================================================================================
====
Epoch 01/3 | train_loss=0.181319 | val_loss=0.127462
Epoch 02/3 | train_loss=0.120076 | val_loss=0.115844
Epoch 03/3 | train_loss=0.109798 | val_loss=0.111792
================================================================================
====
ESM weight decay tuning: wd = 0.0001
================================================================================
====
Epoch 01/3 | train_loss=0.253465 | val_loss=0.186209
Epoch 02/3 | train_loss=0.175128 | val_loss=0.167766
Epoch 03/3 | train_loss=0.159828 | val_loss=0.155417
```

	weight_decay	learning_rate	dropout	best_val_loss	best_micro_f1	best_macro_f1
0	0.00000	0.001	0.3	0.106917	0.159606	0.007095
1	0.00001	0.001	0.3	0.111792	0.148148	0.007200
2	0.00010	0.001	0.3	0.155417	0.138748	0.003919

In [15]:
```
best_wd_row_esm = df_wd_esm.iloc[0]
best_weight_decay_esm = float(best_wd_row_esm["weight_decay"])

print("Best ESM weight decay:", best_weight_decay_esm)
print("Best ESM weight decay row:")
display(best_wd_row_esm.to_frame())
```

```
Best ESM weight decay: 0.0
Best ESM weight decay row:
```

	0
weight_decay	0.0
learning_rate	0.001
dropout	0.3
best_val_loss	0.106917
best_micro_f1	0.159606
best_macro_f1	0.007095
best_threshold_micro	0.75
model_path	esm_wd_0.0.pt

Step 9. Hidden Layer Size Tuning

Why this step?

Hidden layer size controls model capacity.
A model that is too small may underfit, while a model that is too large may overfit.

What we test

We compare several hidden-layer configurations while keeping the previously selected learning rate, dropout, and weight decay fixed.

In [16]:
```
hidden_configs = [
    (256, 128),
    (512, 256),
    (1024, 512)
]

hidden_results_esm = []

for h1, h2 in hidden_configs:
    print("=" * 80)
    print(f"ESM hidden layer tuning: hidden1 = {h1}, hidden2 = {h2}")
    print("=" * 80)

    config = ESM_TUNING_BASE_CONFIG.copy()
    config["learning_rate"] = best_learning_rate_esm
    config["dropout"] = best_dropout_esm
    config["weight_decay"] = best_weight_decay_esm
    config["hidden1"] = h1
    config["hidden2"] = h2
    config["save_prefix"] = f"esm_hidden_{h1}_{h2}"

    result = run_experiment(
        config=config,
```

```
        train_dataset=train_dataset_small,
        val_dataset=val_dataset_small,
        input_dim=input_dim,
        output_dim=output_dim,
        criterion=criterion,
        device=device,
        run_threshold_sweep=True
    )

    hidden_results_esm.append({
        "hidden1": h1,
        "hidden2": h2,
        "learning_rate": best_learning_rate_esm,
        "dropout": best_dropout_esm,
        "weight_decay": best_weight_decay_esm,
        "best_val_loss": result["best_val_loss"],
        "best_micro_f1": result["best_micro_row"]["micro_f1"],
        "best_macro_f1": result["best_micro_row"]["macro_f1"],
        "best_threshold_micro": result["best_micro_row"]["threshold"],
        "model_path": result["model_path"]
    })

df_hidden_esm = pd.DataFrame(hidden_results_esm).sort_values("best_micro_f1"
display(df_hidden_esm)
```

```
==============================================================================
====
ESM hidden layer tuning: hidden1 = 256, hidden2 = 128
==============================================================================
====
Epoch 01/3 | train_loss=0.223920 | val_loss=0.127228
Epoch 02/3 | train_loss=0.119406 | val_loss=0.115267
Epoch 03/3 | train_loss=0.108338 | val_loss=0.110938
==============================================================================
====
ESM hidden layer tuning: hidden1 = 512, hidden2 = 256
==============================================================================
====
Epoch 01/3 | train_loss=0.177665 | val_loss=0.118534
Epoch 02/3 | train_loss=0.111196 | val_loss=0.110919
Epoch 03/3 | train_loss=0.099246 | val_loss=0.107105
==============================================================================
====
ESM hidden layer tuning: hidden1 = 1024, hidden2 = 512
==============================================================================
====
Epoch 01/3 | train_loss=0.154973 | val_loss=0.116322
Epoch 02/3 | train_loss=0.105193 | val_loss=0.109093
Epoch 03/3 | train_loss=0.091802 | val_loss=0.106286
```

	hidden1	hidden2	learning_rate	dropout	weight_decay	best_val_loss	best_micro_
0	1024	512	0.001	0.3	0.0	0.106286	0.15886
1	512	256	0.001	0.3	0.0	0.107105	0.1561
2	256	128	0.001	0.3	0.0	0.110938	0.1459

In [17]:
```
best_hidden_row_esm = df_hidden_esm.iloc[0]
best_hidden1_esm = int(best_hidden_row_esm["hidden1"])
best_hidden2_esm = int(best_hidden_row_esm["hidden2"])

print("Best ESM hidden layer:")
print(f"hidden1 = {best_hidden1_esm}, hidden2 = {best_hidden2_esm}")
print("Best ESM hidden row:")
display(best_hidden_row_esm.to_frame())
```

```
Best ESM hidden layer:
hidden1 = 1024, hidden2 = 512
Best ESM hidden row:
```

	0
hidden1	1024
hidden2	512
learning_rate	0.001
dropout	0.3
weight_decay	0.0
best_val_loss	0.106286
best_micro_f1	0.158863
best_macro_f1	0.009113
best_threshold_micro	0.75
model_path	esm_hidden_1024_512.pt

Step 10. Final Full-Data ESM Model

Why this step?

After tuning on small subsets, we train a final ESM model on the full training and validation split using the selected hyperparameters.

Why this matters

This final stage provides the main model and threshold-based validation results to report.

In [18]:
```
FINAL_CONFIG_ESM = ESM_TUNING_BASE_CONFIG.copy()
FINAL_CONFIG_ESM["learning_rate"] = best_learning_rate_esm
FINAL_CONFIG_ESM["dropout"] = best_dropout_esm
FINAL_CONFIG_ESM["weight_decay"] = best_weight_decay_esm
FINAL_CONFIG_ESM["hidden1"] = best_hidden1_esm
FINAL_CONFIG_ESM["hidden2"] = best_hidden2_esm
FINAL_CONFIG_ESM["num_epochs"] = 5
FINAL_CONFIG_ESM["patience"] = 3
FINAL_CONFIG_ESM["save_prefix"] = "esm_final_full"

print("Final ESM config:")
print(FINAL_CONFIG_ESM)
```

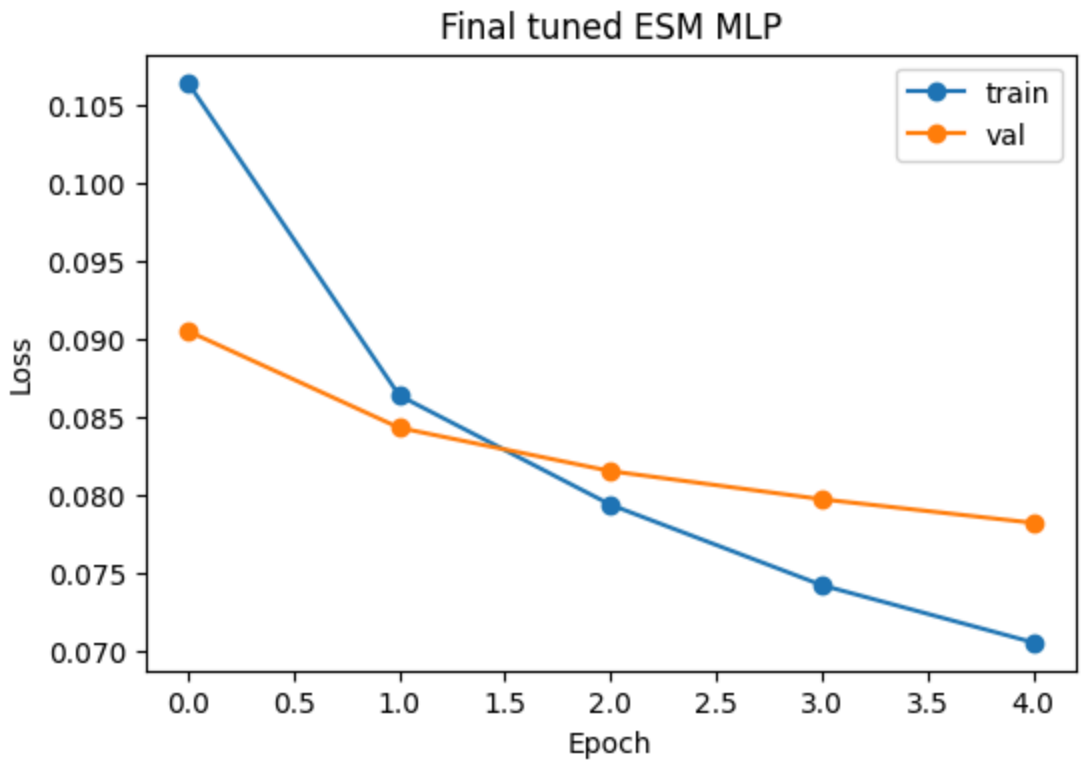
```
Final ESM config:
{'hidden1': 1024, 'hidden2': 512, 'dropout': 0.3, 'learning_rate': 0.001, 'w
eight_decay': 0.0, 'batch_size': 64, 'num_epochs': 5, 'patience': 3, 'save_p
refix': 'esm_final_full'}
```

In [19]:
```
final_result_esm = run_experiment(
    config=FINAL_CONFIG_ESM,
    train_dataset=train_dataset,
    val_dataset=val_dataset,
    input_dim=input_dim,
    output_dim=output_dim,
    criterion=criterion,
    device=device,
    run_threshold_sweep=True
)

print("Final ESM best val loss:", final_result_esm["best_val_loss"])
print("Final ESM best micro-F1:", final_result_esm["best_micro_row"])
print("Final ESM best macro-F1:", final_result_esm["best_macro_row"])
print("Final ESM model path:", final_result_esm["model_path"])
```

```
Epoch 01/5 | train_loss=0.106480 | val_loss=0.090530
Epoch 02/5 | train_loss=0.086398 | val_loss=0.084321
Epoch 03/5 | train_loss=0.079381 | val_loss=0.081538
Epoch 04/5 | train_loss=0.074217 | val_loss=0.079733
Epoch 05/5 | train_loss=0.070544 | val_loss=0.078217
Final ESM best val loss: 0.07821710324761542
Final ESM best micro-F1: {'threshold': 0.7500000000000002, 'micro_f1': 0.201
2154299395264, 'macro_f1': 0.06214149901078674}
Final ESM best macro-F1: {'threshold': 0.7500000000000002, 'micro_f1': 0.201
2154299395264, 'macro_f1': 0.06214149901078674}
Final ESM model path: esm_final_full.pt
```

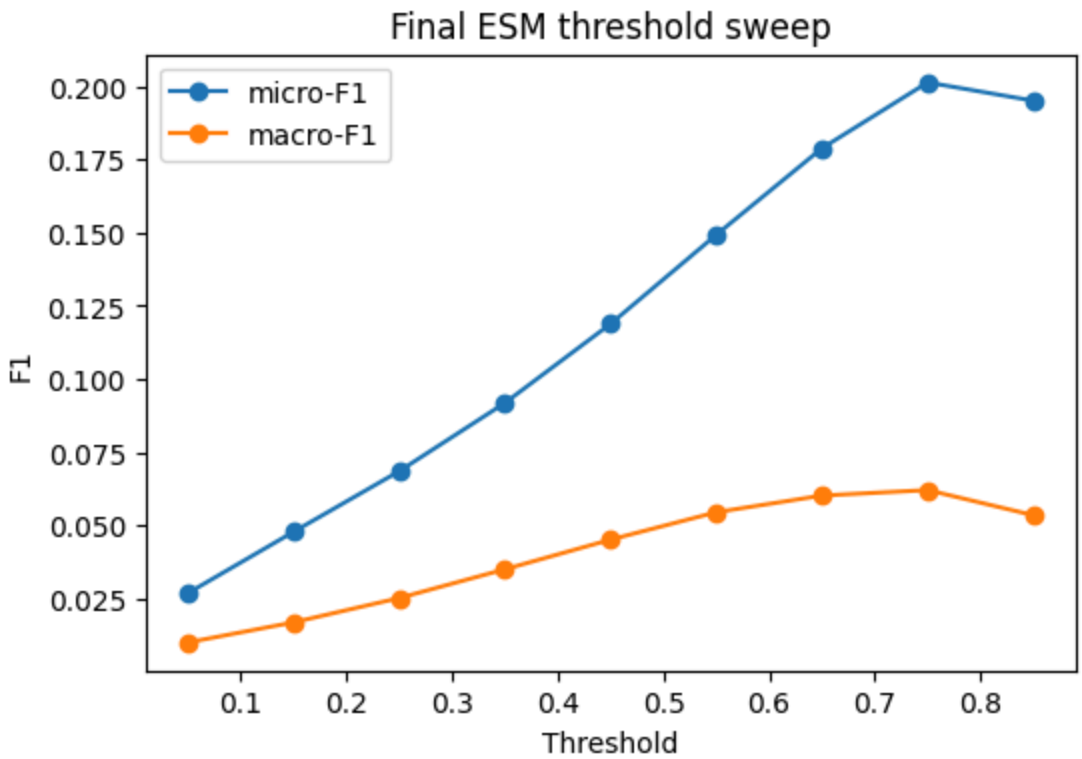
In [20]:
```
plt.figure(figsize=(6, 4))
plt.plot(final_result_esm["train_losses"], marker="o", label="train")
plt.plot(final_result_esm["val_losses"], marker="o", label="val")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Final tuned ESM MLP")
plt.legend()
plt.show()
```



In [21]:
```
df_thresh_esm = pd.DataFrame(final_result_esm["threshold_records"])
display(df_thresh_esm)

plt.figure(figsize=(6, 4))
plt.plot(df_thresh_esm["threshold"], df_thresh_esm["micro_f1"], marker="o",
plt.plot(df_thresh_esm["threshold"], df_thresh_esm["macro_f1"], marker="o",
plt.xlabel("Threshold")
plt.ylabel("F1")
plt.title("Final ESM threshold sweep")
plt.legend()
plt.show()
```

	threshold	micro_f1	macro_f1
0	0.05	0.026964	0.010162
1	0.15	0.047896	0.017013
2	0.25	0.068577	0.025308
3	0.35	0.091867	0.035120
4	0.45	0.118843	0.045235
5	0.55	0.149220	0.054586
6	0.65	0.178755	0.060264
7	0.75	0.201215	0.062141
8	0.85	0.194988	0.053569



Step 11. Save the Selected Hyperparameters and Final Results

Why this step?

Saving the selected settings and final metrics makes the ESM tuning pipeline reproducible and easier to compare with the k-mer notebook.

In [22]:
```
summary_esm = {
    "best_learning_rate": best_learning_rate_esm,
    "best_dropout": best_dropout_esm,
    "best_weight_decay": best_weight_decay_esm,
    "best_hidden1": best_hidden1_esm,
    "best_hidden2": best_hidden2_esm,
    "final_best_val_loss": float(final_result_esm["best_val_loss"]),
    "final_best_micro": final_result_esm["best_micro_row"],
    "final_best_macro": final_result_esm["best_macro_row"],
    "final_model_path": final_result_esm["model_path"],
}

with open("esm_tuning_summary.json", "w") as f:
    json.dump(summary_esm, f, indent=2)

print("Saved tuning summary to esm_tuning_summary.json")
print(summary_esm)
```

```
Saved tuning summary to esm_tuning_summary.json
{'best_learning_rate': 0.001, 'best_dropout': 0.3, 'best_weight_decay': 0.0,
'best_hidden1': 1024, 'best_hidden2': 512, 'final_best_val_loss': 0.07821710
324761542, 'final_best_micro': {'threshold': 0.7500000000000002, 'micro_f1':
0.2012154299395264, 'macro_f1': 0.06214149901078674}, 'final_best_macro':
{'threshold': 0.7500000000000002, 'micro_f1': 0.2012154299395264, 'macro_f
1': 0.06214149901078674}, 'final_model_path': 'esm_final_full.pt'}
```

Summary and Interpretation

What this notebook accomplished

This notebook starts from the original ESM train/validation split strategy and then adopts the small-subset tuning workflow to make hyperparameter search more practical.

What was tuned

We tuned:

- learning rate
- dropout
- weight decay
- hidden layer size

Final outcome

The selected hyperparameters were then transferred to a final full-data ESM model, followed by threshold sweeping on the validation set.

Why this workflow is useful

This preserves consistency with the original ESM pipeline while making tuning substantially more efficient.